Larger frozen-model twist diagnostic

Status: independently replayed finite software-count diagnostic; benchmark admission remains incomplete.

The fixed plan contains 2706 profiles and 2706 attempted executions. There are 60 distinct public DLP targets in this cohort. Configurations and process repetitions are correlated; a recovered execution is not another independent target.

Recorded cohort and outcome	Executions
AA / recovered	6
heldout-reference / recovered	900
heldout-scalar-control / recovered	330
heldout-scalar-control / underfilled	30
heldout / recovered	1050
heldout / underfilled	30
heldout / unsupported-alpha	360

The ratios below use only successful matched recoveries, with every unsuccessful attempt retained in the table above. Endpoints are observed ranges, not confidence intervals.

Numerator / denominator	Median [min, max]; pairs
Full / compressed, cold	0.85955 [0.72674, 0.93308], n=690
Full / compressed, matrix	0.13003 [0.062042, 0.26743], n=690
Ordinary / twist, cold	0.99748 [0.93023, 1.0604], n=660
Map / scalar, cold	0.98548 [0.98099, 0.98928], n=330
IC / strongest completed measured reference	91.231 [58.013, 156.14], n=1380

Ratios below 1 favor the numerator. Compression cost more in every successful matched cold pair in this implementation. The map/scalar saving is available to ordinary bases too. The ordinary/twist ratios do not establish a consistent twist-specific advantage. Five references share each Q: native rho, negation/cycle-escape rho, BSGS, negation BSGS and GP elllog. The strongest completed member is not a global fastest-rho certificate.

Twelve models were fixed by public coefficient/order criteria before their DLP profiles, at minimum16/18/20 subgroup bits. One minimum18 case actually has19 bits. General-model degree17 maps are unsupported and retained as failed method attempts. Original E28 final labels32–63 remain unopened.

All raw instruction sums and completed source/map/full-fiber computations pass independent native/GP checks. Initial GP references were strictly rejected for undeclared automatic stack-growth warnings. Their unchanged-script successors use a larger initial stack and pass the same strict policy. No attack profile was repeated.

The original summary stopped on identical missing query hashes in pre-query failures. A versioned post-data guard correction preserves successful-pair eligibility and formulas. Both versions remain. The pre-data machine protocol also encoded the explicit flat plan in fields the canonical checker treats as a rectangular product. Export schema checks pass, but full protocol/export admission fails (including the analysis-source correction). These are descriptive diagnostics, not promoted benchmark or leaderboard rows. The exact pre-data plan is complete; the checker finding is not evidence of missing planned profiles.

The unit is single-event guest instructions Ir, including actual resident-GP instructions in each exclusive phase. Cold setup, queries, complete decomposition, signs/deduplication, matrix work and source-point verification are charged per target. Input/program loading and result/trace serialization are excluded. No clock, cycle or primitive/GAE conversion is used.

Conditional affine Lean lemmas retain their stated hypotheses. Concrete arithmetic, rank/kernel spans and map fidelity are checked externally; a fully formal concrete bridge is open. Literature and access gaps are in the frozen protocol.

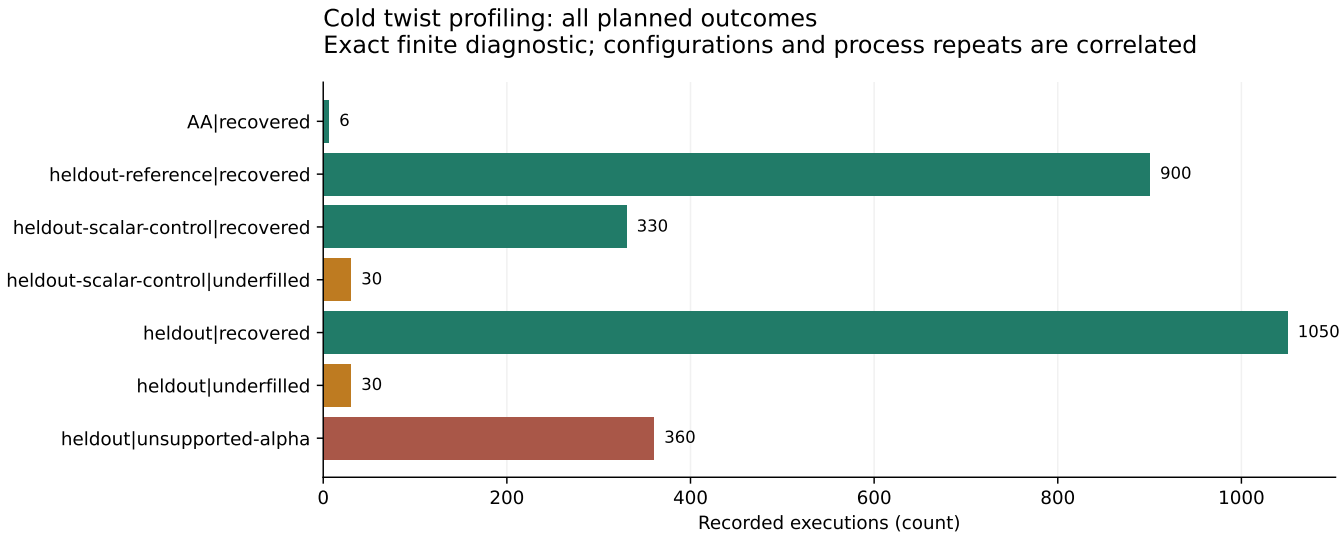


Figure 1: All recorded outcomes

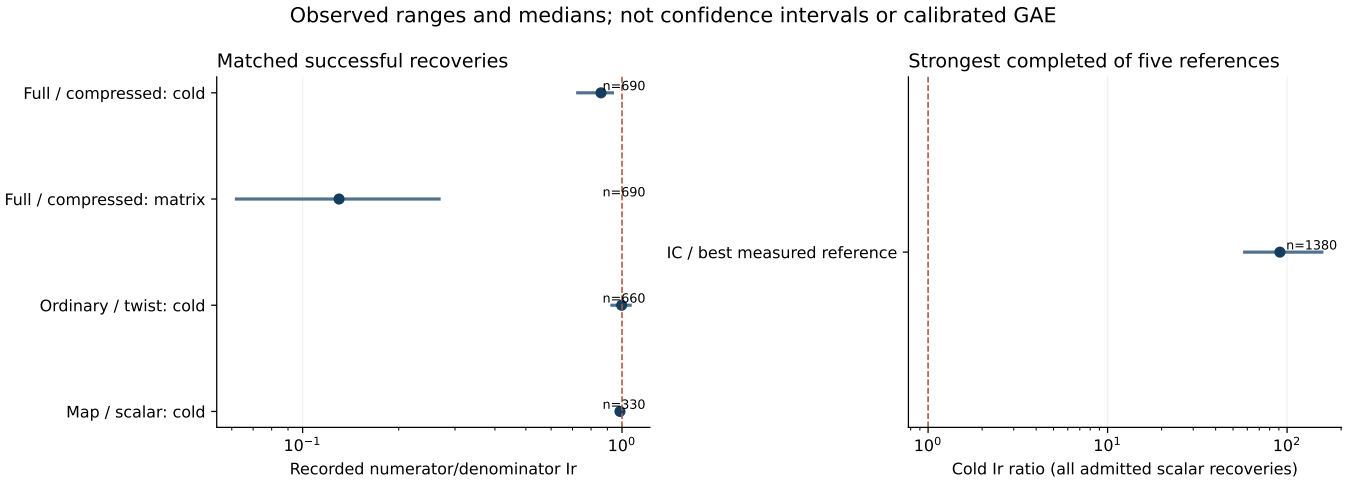


Figure 2: Native recorded ratio ranges

Sources: frozen protocol, native data manifest, source/byte pins. Independent GP admission, reporting correction.

Reproduction and raw-object manifests are linked from the evidence directory. The user-confirmed store is `setec-astronomy-artifacts`, region us-east-2. No NIST, asymptotic, novelty or calibrated-GAE result is established.